

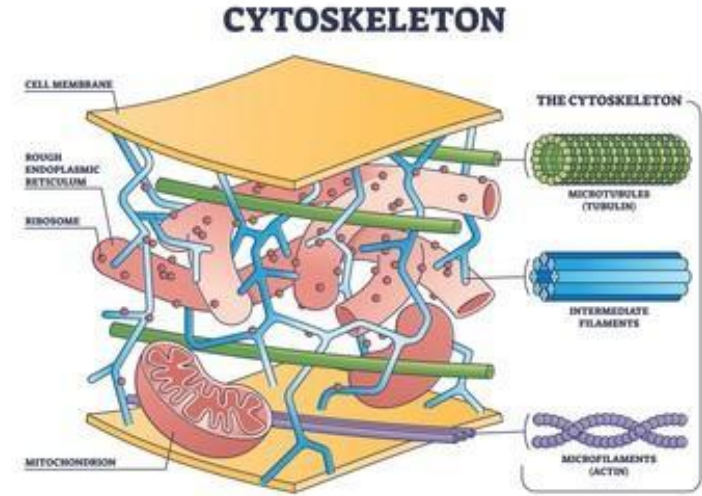
CYTOSKELETON

Structure:

- Threadlike fibers
- Made of proteins

Function:

- Reinforces cell shape
 - Microfilaments: maintains cell shape, muscle contraction
 - Microtubules: Hollow, maintains cell shape and aids in movement of chromosomes during cell division
 - Intermediate Filaments: maintains cell shape, anchorage of nucleus



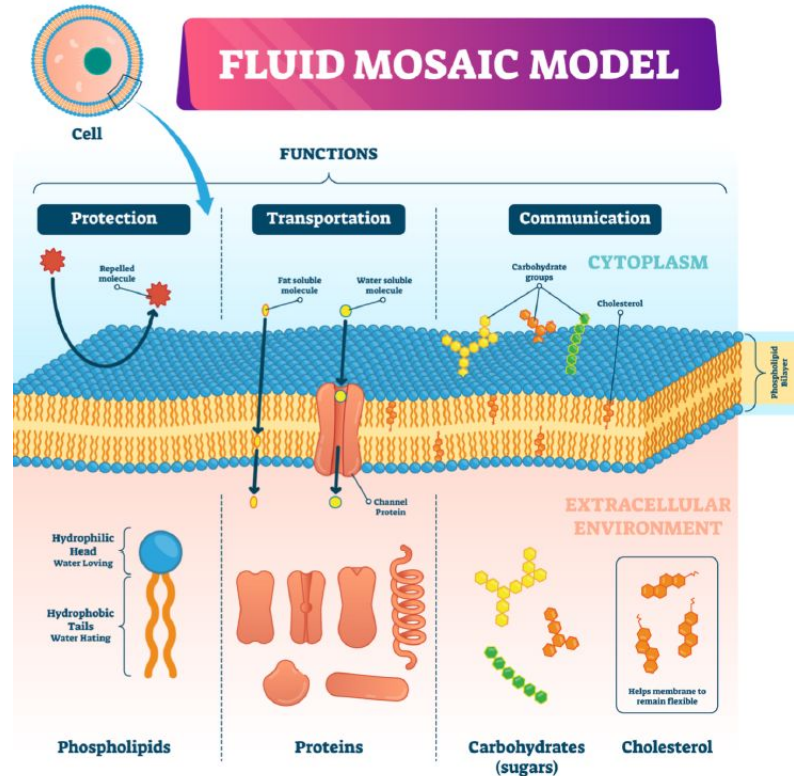
PLASMA MEMBRANE

Structure:

- Semipermeable lipid bilayer studded with proteins and carbohydrates

Function:

- Controls what moves in and out of the cell
- Protects cell
- Found in ALL cells
- Recognizes signals from other cells



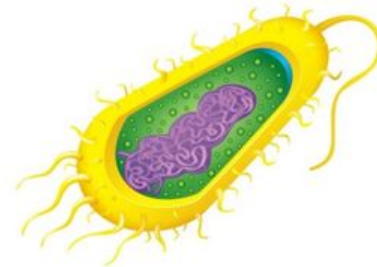
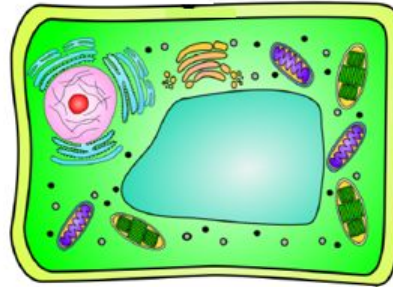
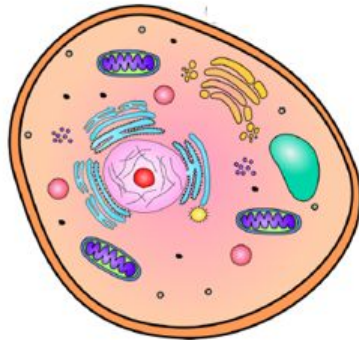
CYTOSOL

Structure:

- Jelly-like substance
- Mainly made up of water
- Cytoplasm = cytosol + organelles suspended in it

Function:

- Holds everything in place
- Provides a solution where chemical reactions take place
- Found in ALL cells



NUCLEUS

Structure:

- Surrounded by a nuclear envelope with pores to control what goes in and out
- Contains genetic material (DNA)
- **NOT** in prokaryotes (bacteria) - they have a non-membrane bound “nucleoid region” in the cytoplasm where DNA is located

Function:

- Protects DNA that controls the activities of the cell

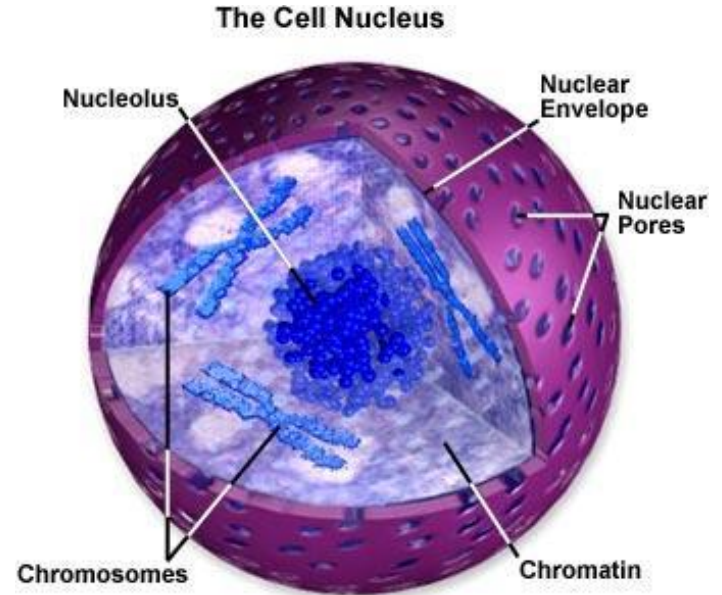


Figure 1

NUCLEOLUS

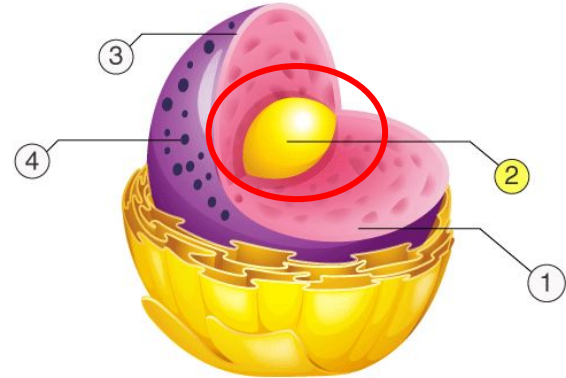
Structure:

- Small structure **inside** of the nucleus

Function:

- Makes a type of RNA that is assembled into ribosomes in the cytoplasm

NUCLEOLUS



1 Nucleoplasm | 2 Nucleolus | 3 Nuclear envelop | 4 Nuclear pore

RIBOSOMES

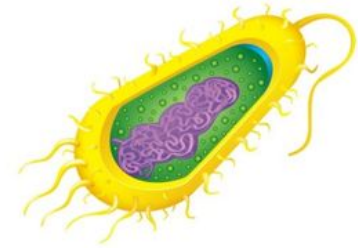
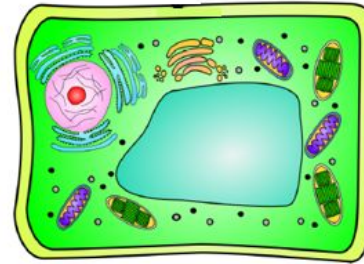
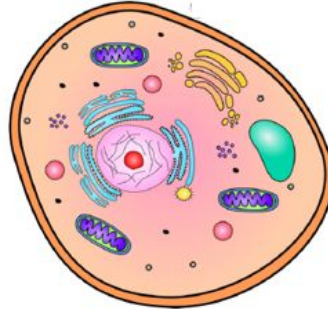
Structure:

- Made of proteins
- Located on the rough ER and floating in cytoplasm
- Large and small subunit
- Are not “membrane-bound”



Function:

- Reads instructions from DNA to make proteins in ALL CELLS
 - Ribosomes on the Rough ER make proteins to export out of the cell
 - Free floating ribosomes make proteins to use within the cell

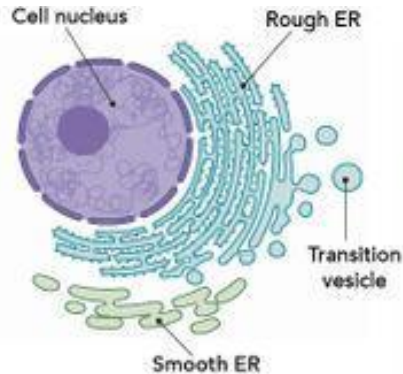


ROUGH ER

(Endoplasmic Reticulum)

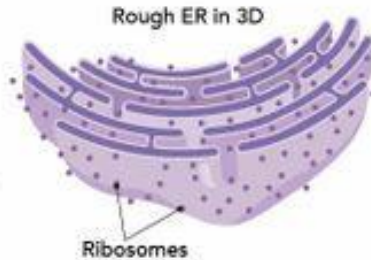
Structure:

- Folded membrane structure that “hugs” the nucleus
- Has ribosomes on the surface



Function:

- Makes and modifies proteins
- Packages proteins for secretion and sends them in vesicles to the Golgi



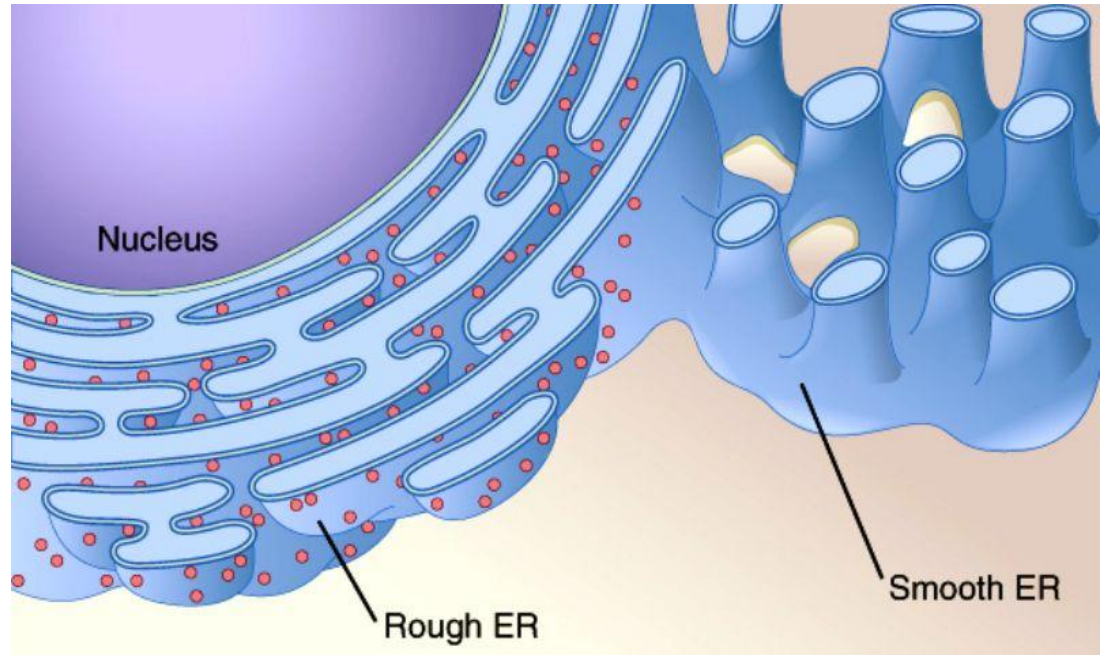
SMOOTH ER (Endoplasmic Reticulum)

Structure:

- Folded membrane structure attached to the rough ER
- NO ribosomes

Function:

- Makes phospholipids and steroid hormones
- Detoxifies (breaks down) harmful chemicals
- Stores calcium



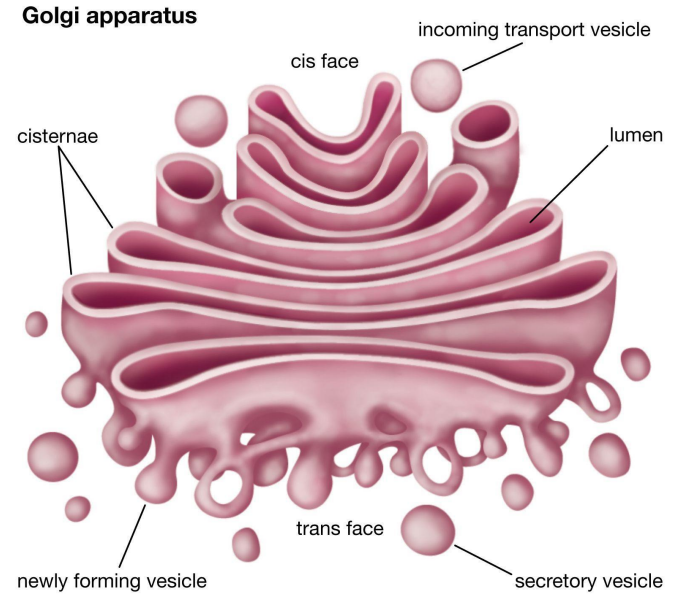
GOLGI APPARATUS (GOLGI BODY)

Structure:

- Folded membrane sacs

Function:

- Receives vesicles of proteins from the ER
- Processes, sorts, and ships proteins where needed
- Produces lysosomes and secretory vesicles



VACUOLES

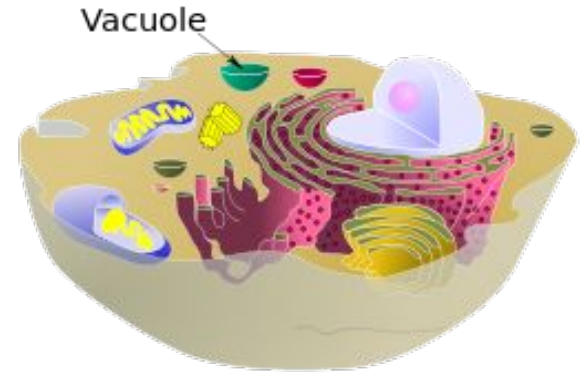
(ANIMAL CELLS)

Structure:

- Membrane bound sacs

Function:

- Storage of water and waste
- Contractile vacuole- found in protists- regulates excess water



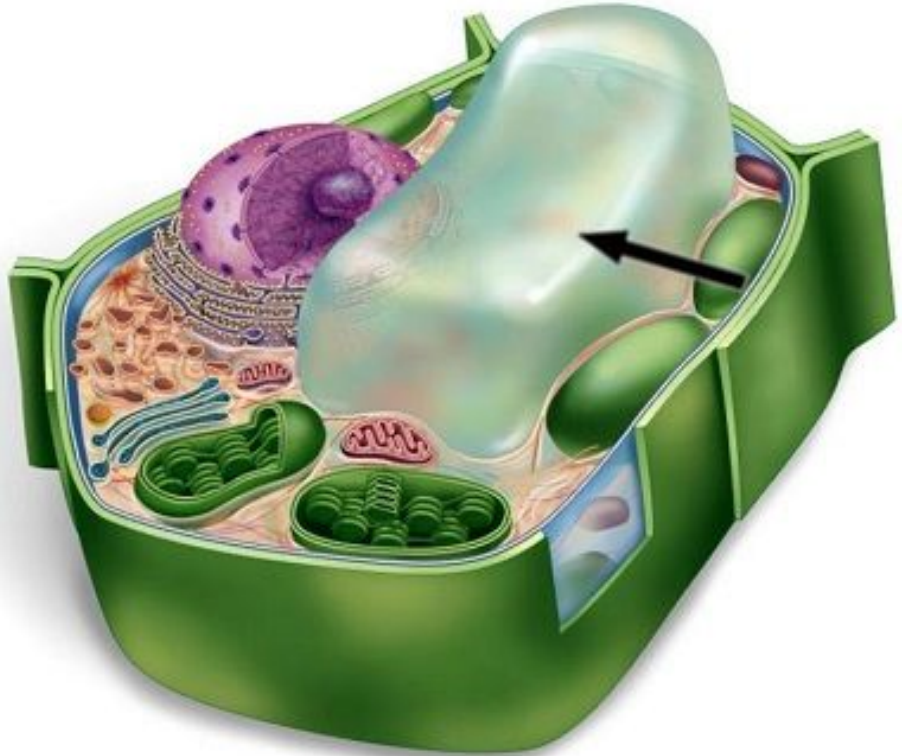
CENTRAL VACUOLE (Plant Cells)

Structure:

- Large and centrally located inside of plant cells

Function:

- Storage of water
- Maintains pressure against the cell wall to help plant cells maintain shape



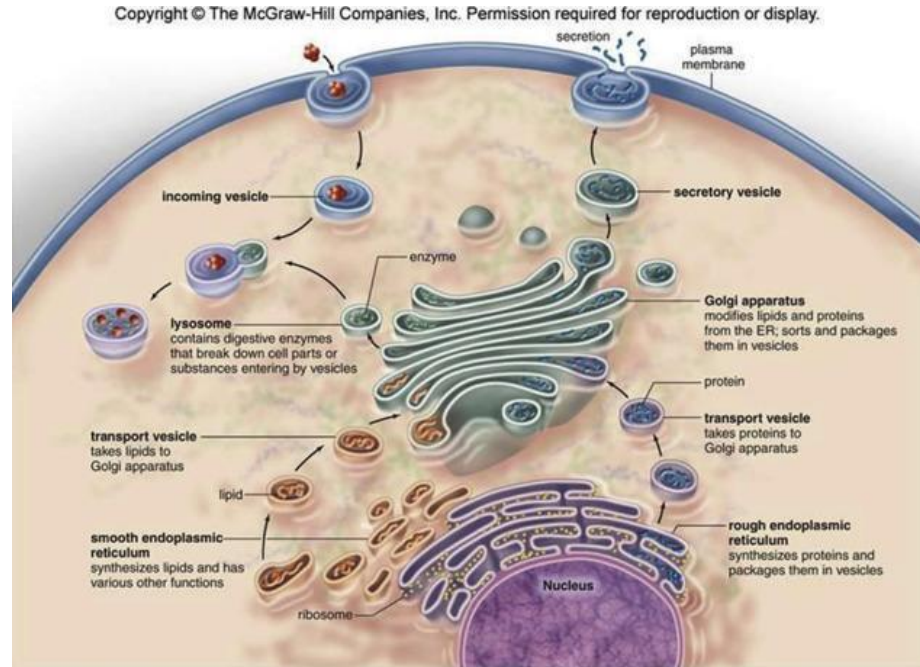
VESICLES

Structure:

- Small sacs made of phospholipids

Function:

- Secretion or transport
- Moves materials around the cell



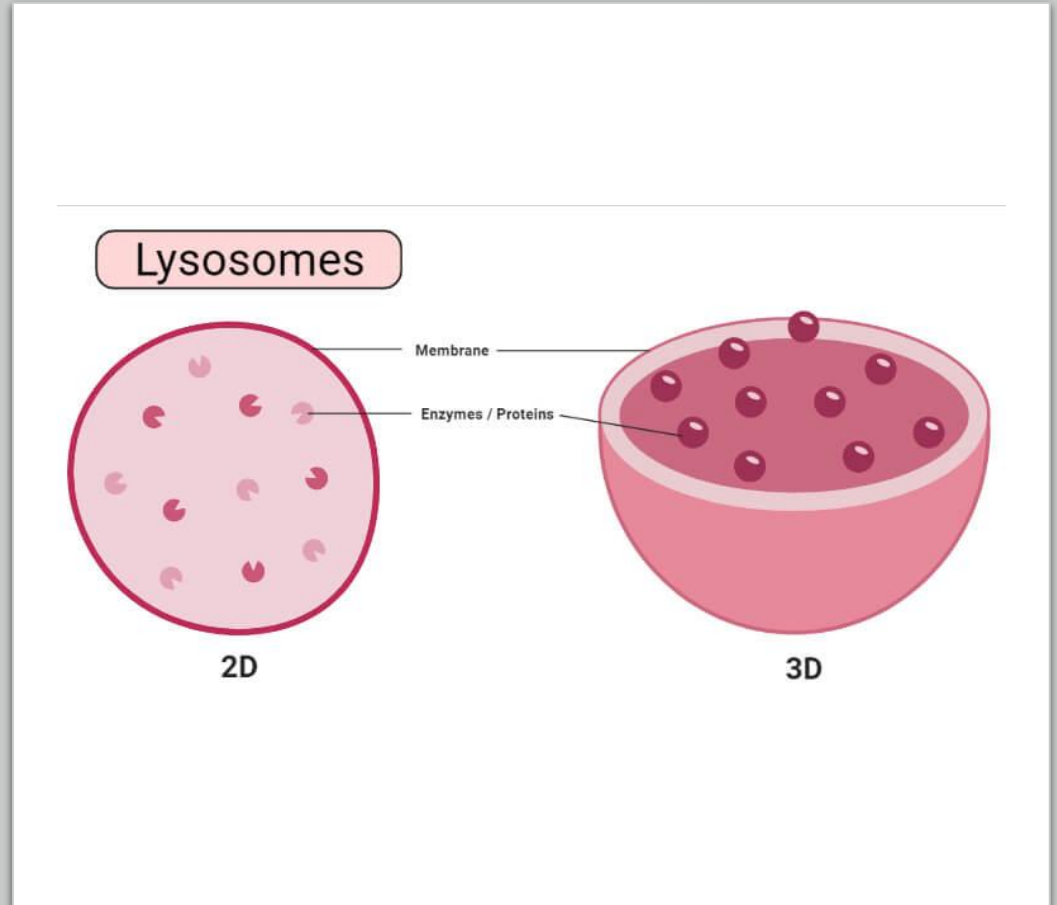
LYSOSOMES

Structure:

- Membrane sac containing enzymes
- ANIMAL CELLS ONLY

Function:

- Break down dead stuff (food, bacteria, old cells parts, etc.)
- Can do programmed cell death (apoptosis) when cell gets old



MITOCHONDRIA

Structure:

- Double membrane
- Has its own DNA
- In BOTH animal and plant cells

Function:

- Makes ATP (energy) by cellular respiration
- Converts glucose to ATP to fuel all cell functions

Mitochondria Structural Features

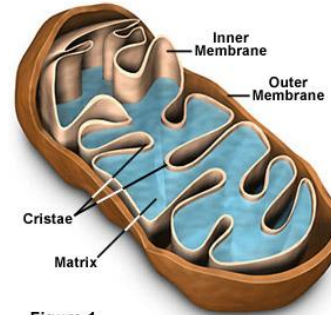
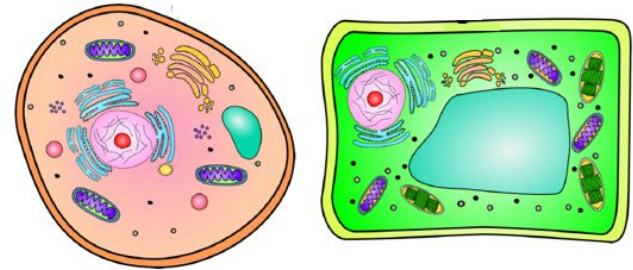


Figure 1



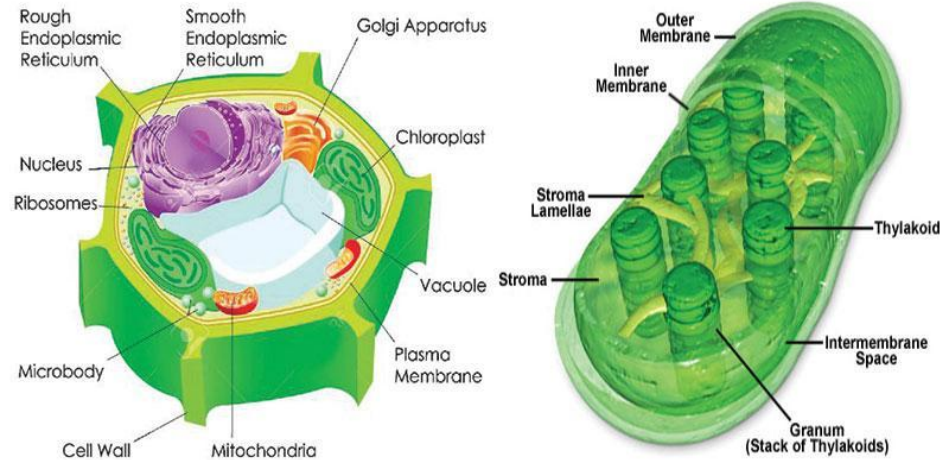
CHLOROPLAST

Structure:

- Double membrane
- Has its own DNA
- PLANT CELLS ONLY

Function:

- Makes sugar and energy using sunlight
 - ATP= active energy
 - Sugar (starches)= stored energy



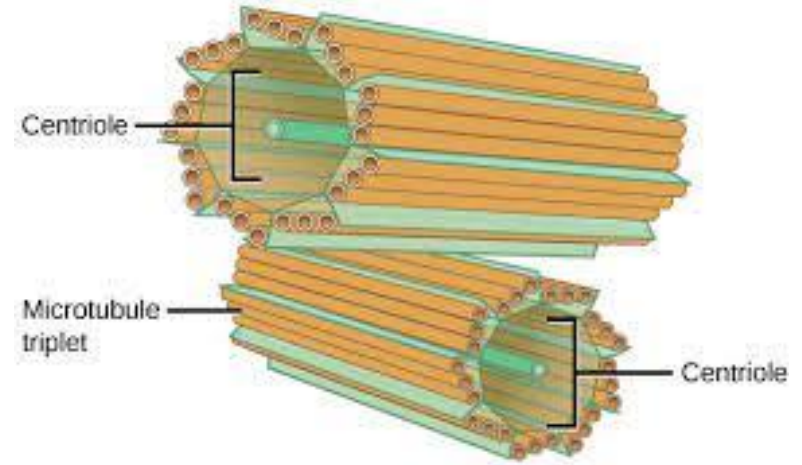
CENTRIOLES

Structure:

- Made of microtubules
- 2 centrioles together= centrosome

Function:

- ANIMAL CELLS ONLY
- Appear during cell division
- Help divide the cell by pulling chromosomes apart



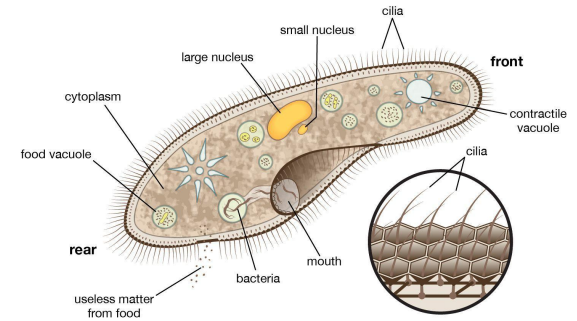
CILIA

Structure:

- Hair-like structures that are extensions of the cell membrane
- Animal and bacteria cells ONLY

Function:

- Help the cell take in food, move, and filter substances



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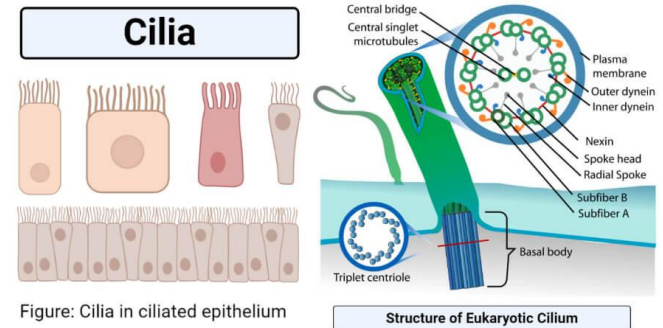


Figure: Cilia in ciliated epithelium

FLAGELLA

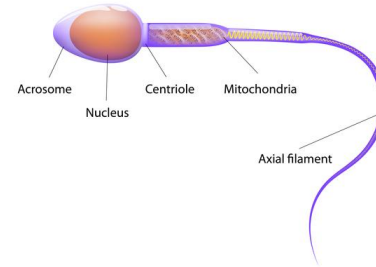
Structure:

- Long whip-like appendage
- Bacteria and animal cells only

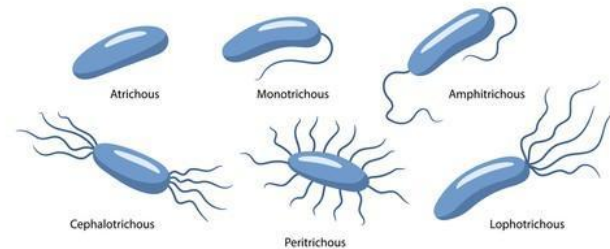
Function:

- Propels or moves the cell
- Found in many protists
- Moves sperm cells in humans

SPERM CELL



Arrangement of bacterial flagelars



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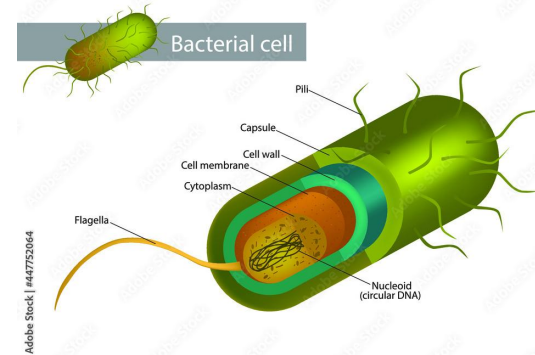
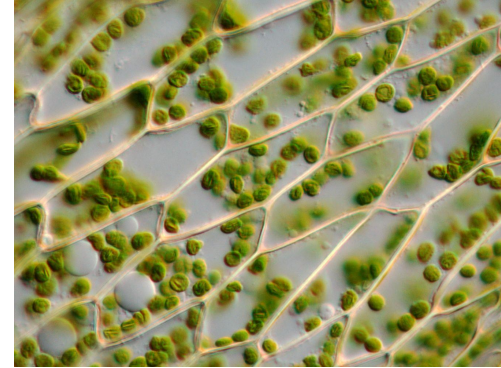
CELL WALL

Structure:

- Surrounds the cell membrane
 - Made of cellulose in plants
 - Made of chitin in fungi
 - Made of peptidoglycan in bacteria
- **All carbohydrates!**

Function:

- Protects the cell and helps it maintain shape
- In plants it prevents cells from expanding too much as water varies in the environment



ENDOMEMBRANE SYSTEM

- In Eukaryotic cells, organelles are connected through the endomembrane system
- Vesicles can bud off from or fuse with other membrane bound organelles which aids in the synthesis, storage, and export of molecules like lipids and proteins.
- Includes endoplasmic reticulum, Golgi apparatus, lysosomes, vacuoles/vesicles, and the plasma membrane (all of these are membrane bound!)

